Evaluating statistical claims: Observational studies and experiments

Problem-Solving and Data Analysis · Math · 12 questions

1 10225-DC · Easy

The members of a city council wanted to assess the opinions of all city residents about converting an open field into a dog park. The council surveyed a sample of 500 city residents who own dogs. The survey showed that the majority of those sampled were in favor of the dog park. Which of the following is true about the city council's survey?

A. It shows that the majority of city residents are in favor of the dog park.

B. The survey sample should have included more residents who are dog owners.

C. The survey sample should have consisted entirely of residents who do not own dogs.

D. The survey sample is biased because it is not representative of all city residents.

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2 05788-DC · Easy

A market researcher selected 200 people at random from a group of people who indicated that they liked a certain book. The 200 people were shown a movie based on the book and then asked whether they liked or disliked the movie. Of those surveyed, 95% said they disliked the movie. Which of the following inferences can appropriately be drawn from this survey result?

A. At least 95% of people who go see movies will dislike this movie.

B. At least 95% of people who read books will dislike this movie.

C. Most people who dislike this book will like this movie.

D. Most people who like this book will dislike this movie.

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3 028921-DC · Medium

Residents of a town were surveyed to determine whether they are satisfied with the concession stand at the local park. A random sample of 200 residents was selected. All 200 responded, and 87% said they are satisfied. Based on this information, which of the following statements must be true?

I. Of all the town residents, 87% would say they are satisfied with the concession stand at the local park.
II. If another random sample of 200 residents were surveyed, 87% would say they are satisfied.

A. Neither

B. I only

C. II only

D. I and II

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4 10199-DC · Medium

A polling agency recently surveyed 1,000 adults who were selected at random from a large city and asked each of the adults, “Are you satisfied with the quality of air in the city?” Of those surveyed, 78 percent responded that they were satisfied with the quality of air in the city. Based on the results of the survey, which of the following statements must be true?

1. Of all adults in the city, 78 percent are satisfied with the quality of air in the city.
2. If another 1,000 adults selected at random from the city were surveyed, 78 percent of them would report they are satisfied with the quality of air in the city.
3. If 1,000 adults selected at random from a different city were surveyed, 78 percent of them would report they are satisfied with the quality of air in the city.

A. None

B. II only

C. I and II only

D. I and III only

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5 029581-DC · Medium

A survey was conducted using a sample of history professors selected at random from the California State Universities. The professors surveyed were asked to name the publishers of their current texts. What is the largest population to which the results of the survey can be generalized?

A. All professors in the United States

B. All history professors in the United States

C. All history professors at all California State Universities

D. All professors at all California State Universities

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6 029003-DC · Hard

A sample of 40 fourth-grade students was selected at random from a certain school. The 40 students completed a survey about the morning announcements, and 32 thought the announcements were helpful. Which of the following is the largest population to which the results of the survey can be applied?

A. The 40 students who were surveyed

B. All fourth-grade students at the school

C. All students at the school

D. All fourth-grade students in the county in which the school is located

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7 11044-DC · Hard

To determine the mean number of children per household in a community, Tabitha surveyed 20 families at a playground. For the 20 families surveyed, the mean number of children per household was 2.4. Which of the following statements must be true?

A. The mean number of children per household in the community is 2.4.

B. A determination about the mean number of children per household in the community should not be made because the sample size is too small.

C. The sampling method is flawed and may produce a biased estimate of the mean number of children per household in the community.

D. The sampling method is not flawed and is likely to produce an unbiased estimate of the mean number of children per household in the community.

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8 4c42a273-cc25-495c-8315-77ecc0dd2ea9 · Hard

For a baobab tree habitat in South Africa, a scientist randomly selected 50 baobab trees that were 17 years old and randomly assigned them to two groups. Each group was subjected to a different watering pattern for 2 consecutive years to observe whether the watering pattern affects the trees' growth rate. Based on the design of the study, what is the largest group to which these results can be applied?

A. All the 50 baobab trees that were selected in this habitat

B. All the baobab trees that were 19 years old in this habitat

C. All the baobab trees that were 17 years old in South Africa

D. All the baobab trees that were 17 years old in this habitat

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9 029344-DC · Hard

A psychologist designed and conducted a study to determine whether playing a certain educational game increases middle school students' accuracy in adding fractions. For the study, the psychologist chose a random sample of 35 students from all of the students at one of the middle schools in a large city. The psychologist found that students who played the game showed significant improvement in accuracy when adding fractions. What is the largest group to which the results of the study can be generalized?

A. The 35 students in the sample

B. All students at the school

C. All middle school students in the city

D. All students in the city

10 86e6dfb4-d2c8-45de-afe3-b133ed0d88a5 · Hard

A trivia tournament organizer wanted to study the relationship between the number of points a team scores in a trivia round and the number of hours that a team practices each week. For the study, the organizer selected 55 teams at random from all trivia teams in a certain tournament. The table displays the information for the 40 teams in the sample that practiced for at least 3 hours per week.

Hours practiced	Number of points per round		
	6 to 13 points	14 or more points	Total
3 to 5 hours	6	4	10
More than 5 hours	4	26	30
Total	10	30	40

Which of the following is the largest population to which the results of the study can be generalized?

A. All trivia teams in the tournament that scored 14 or more points in the round

B. The 55 trivia teams in the sample

C. The 40 trivia teams in the sample that practiced for at least 3 hours per week

D. All trivia teams in the tournament

11 020744-DC · Hard

Near the end of a US cable news show, the host invited viewers to respond to a poll on the show's website that asked, "Do you support the new federal policy discussed during the show?" At the end of the show, the host reported that 28% responded "Yes," and 70% responded "No." Which of the following best explains why the results are unlikely to represent the sentiments of the population of the United States?

A. The percentages do not add up to 100%, so any possible conclusions from the poll are invalid.

B. Those who responded to the poll were not a random sample of the population of the United States.

C. There were not 50% "Yes" responses and 50% "No" responses.

D. The show did not allow viewers enough time to respond to the poll.

12 5d299984-b0c3-416d-b959-03fecfb8571e · Hard

A statistician conducted a study to investigate the relationship between the average number of minutes piano students practiced each week and the percentage of correct notes the students played during a recital. For the study, the statistician selected a sample of 85 piano students at random from all the piano students in a certain region who were in their second year with their conservatory. The table shows the information for 75 piano students in this sample who practiced at least 50 minutes on average each week.

	Percentage of correct notes the students played during a recital			
Average number of minutes practiced each week	Less than 50%	50% to 80%	Greater than 80%	Total
50 to 150	7	28	5	40
151 or more	3	11	21	35
Total	10	39	26	75

Which of the following is the largest population to which the results of the study can be generalized?

A. All piano students who were in their second year with their conservatory

B. All piano students in the region

C. All piano students in the region who were in their second year with their conservatory

D. The 75 piano students in the sample who practiced at least 50 minutes on average each week